

STAGE ONE EDUCATION

Workshop Guide

Robotics Drone - Mechanical Engineering - Artificial Intelligence

ROBOTICS DRONE WORKSHOP

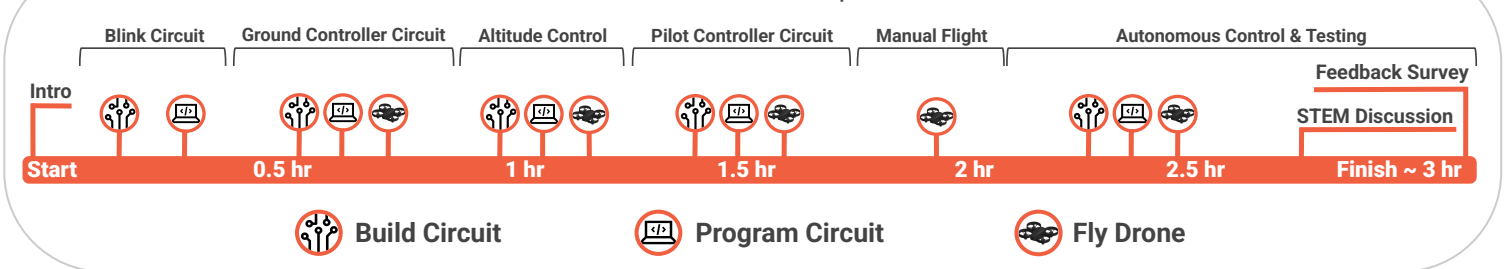
Electronics & Coding

In the Stage One Robotics Workshop, students get a drone, but **no controller!**

Learn engineering by building your own RC controller for a quadrotor:

- Wire a complete circuit, including a **dual core microcontroller** with Wi-Fi, 6-axis accelerometer, lights, potentiometers, and buttons.
- With your team, learn how to **read and translate circuit diagrams** into real, working circuits.
- **Program, upload, and execute** your own autonomous flights.
- Participate in the autonomous airshow to showcase your custom flight path.

Drone Robotics Workshop Timeline



This was **extremely well done**. The instructors were **very nice and approachable**. I am extremely grateful for this as it has **opened my eyes** to a new area of electronics I never would've thought I would do.

- Student San Francisco, CA 2025

"Thank you for going around and **checking on everyone** keep doing that"

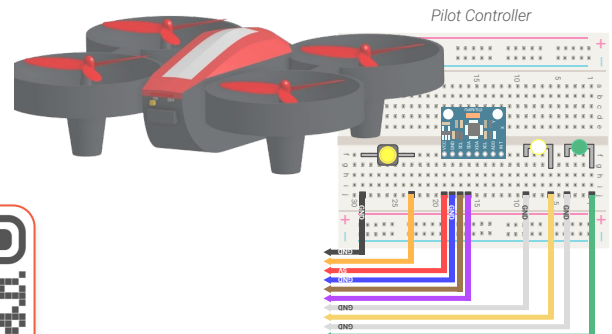
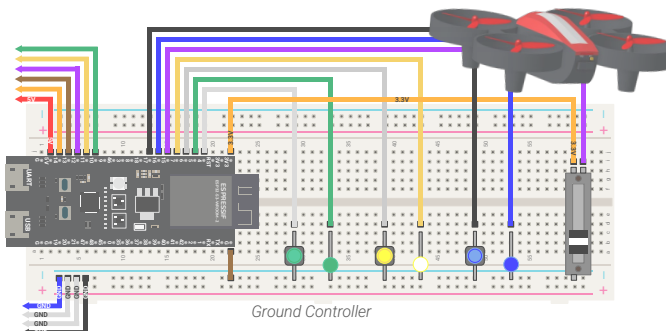
- Student Atlanta, GA 2024

"This was the **best workshop in the whole program**"

- Student Boston, MA 2024

Yeeshaw-tastic

- Student Houston, TX 2025

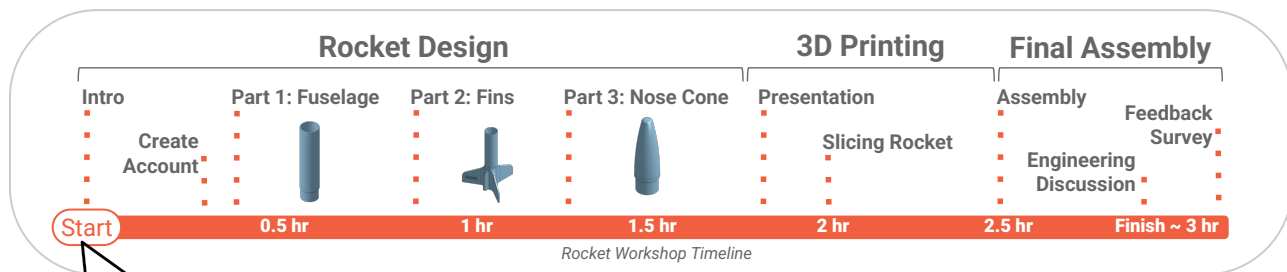


<https://stageoneeducation.com/robotics-workshop.html>

MECHANICAL ENGINEERING WORKSHOP

In the Stage One Mechanical Engineering Workshop, students **become rocket designers!**

- Design and engineer your own three-part rocket using industry-standard Computer Aided Design (CAD) software.
- Guided by **GIF-style micro-videos** and concise text, no prior experience is required.
- Master core CAD skills: sketching, parametric modeling, assemblies.
- Live 3D-printing demo.
- Learn from open-forum Q&A where you can ask our **real engineers** anything about your design, CAD techniques, and careers in engineering.



"I loved this workshop. I've never used any CAD program before and I was able to learn how and finish the entire rocket within the allotted time. I'm glad I know how to use programs like this and would like to try with a little less guidance next time. :)

- Student Los Angeles, CA 2025

"Great workshop, I had an awesome time while learning valuable skills that will help essential for my success later in engineering and beyond.

- Student Atlanta, GA 2025

"The guidance coupled with creative freedom of the workshop was perfect!

- Student Berkeley, CA 2025



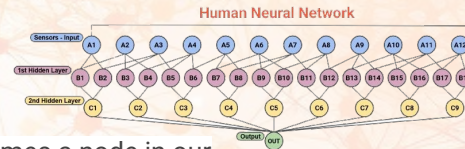
"I really enjoyed this workshop! Thank you for giving me this experience, I am certain that it is a stepping stone into my future.

- Student Houston, TX 2024

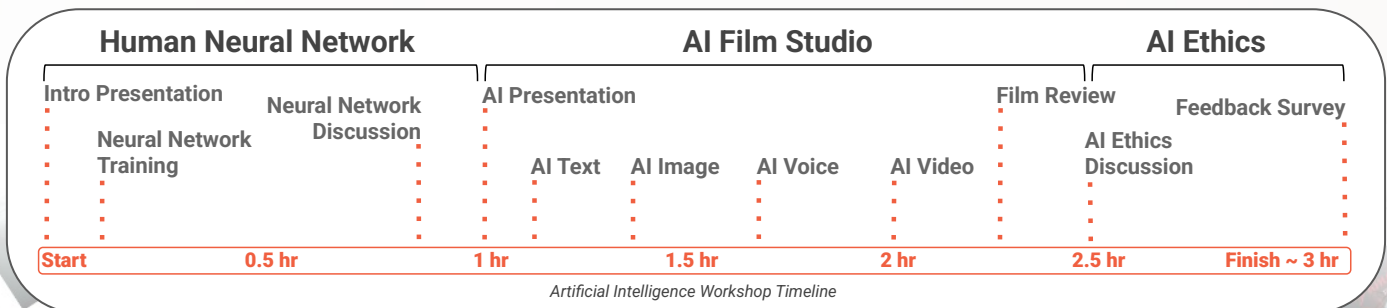
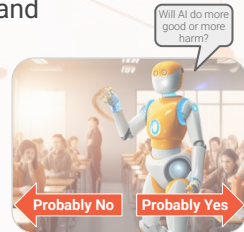
<https://stageoneeducation.com/rocket-workshop.html>

ARTIFICIAL INTELLIGENCE WORKSHOP

In the Stage One AI Workshop, participants train a human neural network, explore advanced AI models, and discuss the ethics of AI through an unforgettable, **hands on experience**.



- **Human Neural Network**
 - Join an interactive game of telephone where each participant becomes a node in our **classroom-sized neural network**.
 - Gain an **intuitive grasp** of neural networks and their learning processes—no math and no computers.
- **AI Film Studio**
 - Your movie needs a script, a film poster, trailer, voice acting, and a budget. You've got 45 minutes.
 - Build everything with the **most advanced AI models** – text, image, video generation, and clone your own voice for the main character.
- **AI Ethics**
 - **Do you own your film?** Whose name goes on the credit roll?
 - Congratulations – your film made \$500,000,000! **Who gets paid?** You? Us? OpenAI?



Movie posters created by participants using the leading AI models

I was truly moved by this workshop, and I think it will be **the most memorable activity of my school trip**. I could really feel the difference in the quality of education compared to my home country. If I ever get the chance, I would love to participate again.

- Student Boston, MA 25'

One of the **best AI workshops** I've ever done. I love how simple the AI tools were.

- Student Houston, TX 24'

It was really great. I first thought that I would not understand it but you did such a good thing instruction for us. **I would never forget this experience!!!** Thank you so much 😊

- Student Boston, MA 25'

It was **super fun**, learning how to utilize AI more and finding out how much AI can actually do.

- Student Atlanta, GA 25'

Liked the social activity and instructors simpleness yet effective in explaining information

- Student Washington, D.C. 24'



Thank you for answering our questions! This has been the first place that I've been able to **fully understand neural networks** :)

- Student Berkeley, CA 25'

<https://stageoneeducation.com/ai-workshop.html>